

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

JEE(Main)**UNIT TEST****19-09-2021****JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, I, I(B), II(A), II & NURTURE (LIVE) I, I(B), II(A)]**

Time : 3 Hours

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY**Important Instructions :**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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SECTION-I : (Maximum Marks: 80)

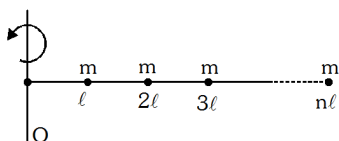
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

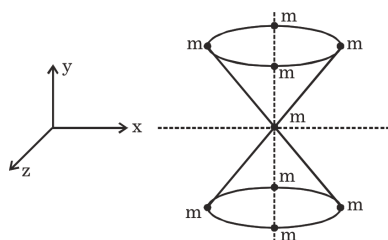
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Find moment of inertia about axis OA

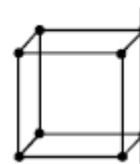


- (A) $\frac{m\ell^2}{6}$
 (B) $\frac{m\ell^2}{6}n(n+1)(2n+1)$
 (C) zero
 (D) $m\ell^2 \frac{n(n+1)}{2}$
2. Nine point mass m are arranged in an double-cone system as shown in diagram. Radius of each cone is R and height of each cone is H . The moment of inertia about the x -axis passing through the common vertex is :

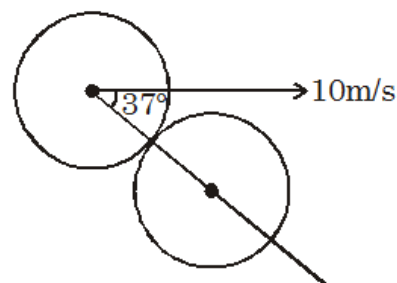


- (A) $8mH^2$ (B) $8m(H^2 + R^2)$
 (C) $8mH^2 + 4mR^2$ (D) None of these

3. Eight point objects, each of mass M , are at the vertices of a rigid, massless framework in the form of a cube; each side is of length a . What is the moment of inertia of this arrangement about an axis passing through one edge of the cube?



- (A) $4Ma^2$ (B) $6Ma^2$ (C) $8Ma^2$ (D) $10Ma^2$
4. A bullet is fired from a gun. The force on the bullet is given by $F = 600 - (2 \times 10^5)t$, where F is in Newton and t in second. The force on the bullet becomes zero as soon as it leaves the barrel. The impulse imparted to the bullet is
 (A) 8 N-s (B) 0.9 N-s
 (C) 1.8 N-s (D) 2.4 N-s
5. A ball of mass m collides another identical ball at rest obliquely as in figure. If velocity of 2nd ball after the collision is $6m/s$, then the value of coefficient of restitution is :

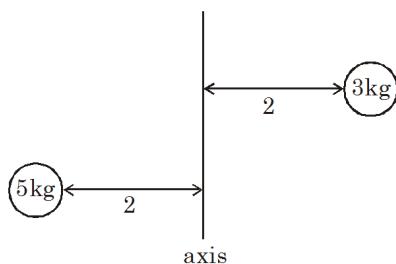


- (A) 1 (B) $\frac{1}{2}$
 (C) $\frac{3}{2}$ (D) $\frac{1}{4}$

6. A hemispherical shell of radius R is capped on a vertical hollow cone of base radius R and height $\frac{3R}{4}$ (as in an ice cream cone). Both the cone and the shell have the same mass per unit area. Find the distance of the center of mass of the system from the vertex of the cone.

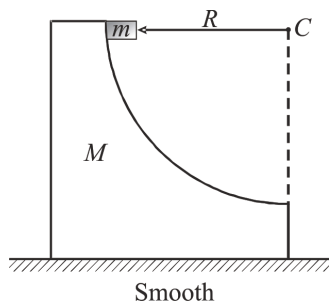
(A) $\frac{35R}{46}$ (B) $\frac{23R}{28}$ (C) $\frac{18R}{25}$ (D) $\frac{25R}{26}$

7. As shown in the figure moment of inertia of system is :



(A) 32 kg-m^2 (B) 30 kg-m^2
(C) 12 kg-m^2 (D) 20 kg-m^2

8. A small mass m is placed and released from the vertical portion of a quarter circular arc as shown in figure. As it slides, the acceleration of the mass M on the smooth horizontal surface:

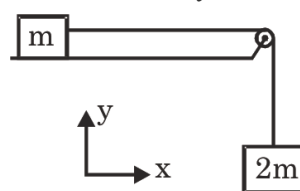


- (A) Continuously increases
(B) Continuously decreases
(C) First increases then decreases
(D) First decreases then increases

9. A particle of mass $2m$ is projected at an angle of 45° with horizontal with a velocity of $20\sqrt{2} \text{ m/s}$. After 1 sec due to explosion, the particle breaks into two equal pieces. As a result of this one part comes to rest. The maximum height from ground attained by the other part is ($g = 10 \text{ m/s}^2$)

(A) 25 (B) 50 m
(C) 40 m (D) 35 m

10. In the given figure table is smooth and pulley and string are ideal then acceleration of centre of mass of the two blocks system is :-

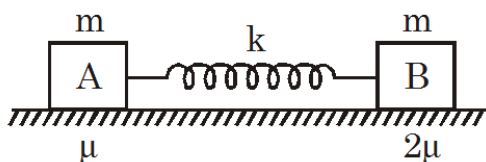


(A) $\frac{g}{2} \hat{i} - \frac{g}{2} \hat{j}$ (B) $\frac{2g}{9} \hat{i} - \frac{4g}{9} \hat{j}$
(C) $-g \hat{j}$ (D) $\frac{g}{2} \hat{i} - g \hat{j}$

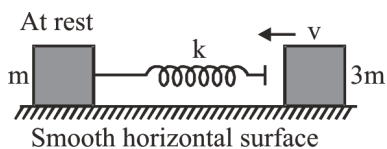
11. A small car is moving along a straight level highway at a speed of $3v$. The car hits from behind a large truck moving in the same direction at speed v . After the collision, the car is stuck onto the truck. During the collision, which vehicle experiences the greater average forces (magnitude) ?

- (A) The car
(B) The truck
(C) The forces are equal
(D) Impossible to determine without information about the masses

12. Two block of same mass 'm' are joined with an ideal spring of spring constant k and kept on a rough surface as shown in figure. The spring is initially unstretched and coefficient of friction between the blocks and horizontal surface is μ and 2μ respectively. What should be the maximum speed of block 'A' such that the block 'B' does not move ?



- (A) $2\mu g \sqrt{\frac{2m}{k}}$ (B) $\mu g \sqrt{\frac{2m}{k}}$
 (C) $\mu g \sqrt{\frac{m}{k}}$ (D) $2\mu g \sqrt{\frac{m}{2k}}$
13. Maximum compression in the spring (light) in the physical situation shown is :-

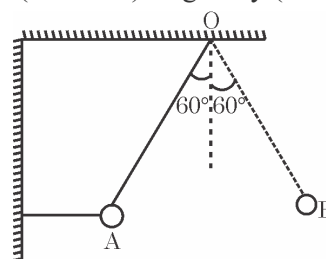


- (A) $\sqrt{\frac{3m}{2k}} v$ (B) $\sqrt{\frac{3m}{k}} v$
 (C) $\sqrt{\frac{3m}{16k}} v$ (D) $\sqrt{\frac{3m}{4k}} v$
14. A particle of mass m_1 collides head on with a stationary particle of mass m_2 . If $\frac{m_1}{m_2} > e$, where e is the coefficient of restitution, then:
- (A) m_1 will return back
 (B) m_1 will move in same direction
 (C) m_1 will stop
 (D) unpredictable

15. A ball is thrown vertically upwards from the ground. Work done by air resistance during its time of flight is

- (A) Positive during ascent and negative during descent
 (B) Positive during ascent and descent
 (C) Negative during ascent and positive during descent
 (D) Negative during ascent and descent

16. A ball is held at rest in position A by two light cords. The horizontal cord is now cut and ball can swing in vertical plane. What is the ratio of the tension in cord OB in position B to that in position A (cord OA) originally (Before cut) ?



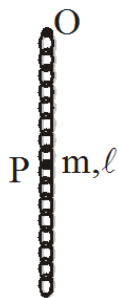
- (A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) 1 (D) $\frac{1}{4}$
17. A machine delivers power to a body which is proportional to velocity of the body. If the body starts with a velocity which is almost negligible; then distance covered by the body is proportional to :

- (A) $\sqrt{v}$
 (B) $\sqrt[3]{\frac{v}{2}}$
 (C) $v^{5/3}$
 (D) v^2

18. The breaking tension of a string is 10N. A particle of mass 0.1 kg tied to it is rotated along a horizontal circle of radius 0.5 meter on a smooth horizontal surface. The maximum speed with which particle can be rotated is :-

(A) $\sqrt{5}$ m/s (B) $\sqrt{50}$ m/s
(C) $\sqrt{500}$ m/s (D) $\sqrt{1000}$ m/s

19. A chain of mass m & length ℓ is suspended from a point O. Now, the midpoint P of the chain is also taken & slowly put at point O. Find the work done by gravity in this process.



(A) $\frac{3mg\ell}{16}$ (B) $\frac{9mg\ell}{40}$
(C) $\frac{5mg\ell}{16}$ (D) $\frac{7mg\ell}{16}$

20. An object starts moving from rest on a circle of radius $2m$, with constant tangential acceleration $2m/s^2$. The time after which acceleration of object makes angle 45° from tangent is

(A) 1 sec (B) 2 sec
(C) $\sqrt{2}$ sec (D) $\frac{1}{\sqrt{2}}$ sec

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

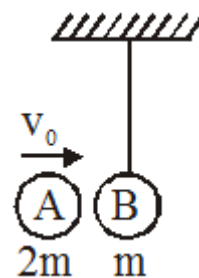
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

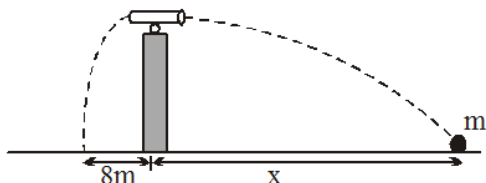
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

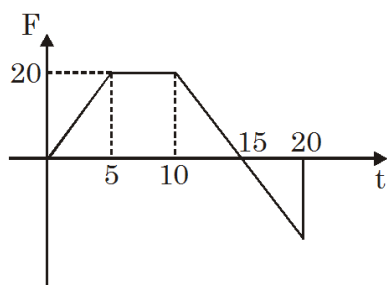
- Three point masses each of mass 2 kg are placed at the corners of an equilateral triangle of side $\sqrt{3}m$. Then the moment of inertia of this system about an axis passing along one side of the triangle is (In $Kg\cdot m^2$)
- A sphere A of mass 2kg moving with velocity v_0 collide with bob of mass 1kg as shown in the figure. value of tension at which string of pendulum break is 40N. If length of pendulum is 30cm, then find the maximum value of v_0 in m/s so that string does not break. Coefficient of restitution is $\frac{1}{2}$.



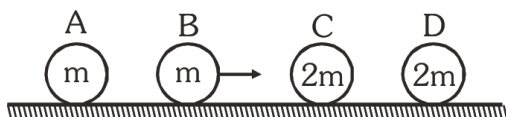
3. A cannon of mass $M = 200 \text{ kg}$ is positioned at the top of a narrow wall as shown. It fires a ball of mass $m = 2 \text{ kg}$ horizontally across a plane. Unfortunately the gunners forgot to lock the frictionless wheels of the cannon and it immediately rolls backwards off the wall, landing a distance 8 m from the wall as shown. Neglecting air friction, at what distance x (in meters) from the base of the wall does the ball land?



4. $F-t$ graph of a particle starts from rest is shown. At what time (in sec), linear momentum of the particle is maximum?



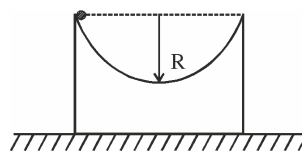
5. Four balls A, B, C and D are placed on a smooth horizontal surface. Ball B collides with ball C with an initial velocity u as shown in figure. If all collisions are elastic, find the total number of collisions between the balls.



6. Block A of mass 0.50 kg initially travelling towards right at 4.00 m/s on a frictionless horizontal track collides elastically with another block B of mass 1.50 kg initially at rest. If the collision is head on, find magnitude of momentum in SI units of the block B after the collision.



7. A particle of mass m is released from the top of a wedge. Mass of wedge is $4m$. All the surfaces are assumed to be frictionless and radius of circle $R = 16 \text{ m}$. Then find the speed of the particle when it reaches the lowest point.



8. A man weighing 80 kg is standing on a trolley weighing 320 kg . The trolley is resting on frictionless horizontal rails. If the man starts walking on the trolley along the rails at a relative speed of 1 m/s , then after 4 s his displacement relative to the ground will be (in m).
9. An object of mass 4 kg falls from rest through a vertical distance of 10 m and reaches the ground with a speed of 14 m/s . How much work is done against air friction in joules? [$g = 10 \text{ m/s}^2$]
10. An escalator is used to move 5 people (60 kg each) per minute from the first floor of a department store to the second floor, 20 m above. Neglecting friction, calculate the power (in kW) required.

SECTION-I : (Maximum Marks: 80)

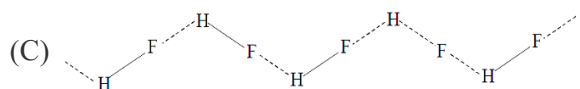
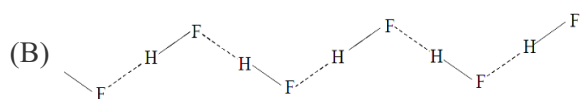
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Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Which of the following is the correct representation of H-bonding in HF (Solid)?



(D) All of these

2. Atom Y is found to be twelve times as heavy as an atom of X. If one wished to prepare a compound in which 7.2 gm of Y would be required to combine with 1 gm of X. What is the simplest formula of compound ?



3. 0.1 mole H_3PO_x is completely neutralised by 5.6 g KOH then the true statement is:

- (A) $x = 3$ and given acid is dibasic
(B) $x = 4$ and given acid has no P-H linkage
(C) $x = 2$ and given acid has two P-H linkage
(D) All of these

4. Which of the following groups of molecules act both as oxidising agent as well as reducing agent?

- (A) $KMnO_4$, O_3 , SO_3
(B) $HClO_4$, HNO_2 , H_2O_2
(C) HNO_2 , SO_2 , H_2O_2
(D) HNO_3 , SO_2 , H_2SO_4

5. The hardness of a 1 litre water sample (in terms of equivalents of $CaCO_3$) containing 1.11 mg $CaCl_2$ is :

- (A) 10 ppm
(B) 5 ppm
(C) 90 ppm
(D) 1 ppm

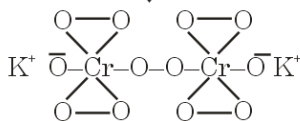
6. 1 mole of ferric oxalate $[Fe_2(C_2O_4)_3]$ is oxidised by x mole of MnO_4^- in strong basic medium and 1 mole of ferrous oxalate $[FeC_2O_4]$ is oxidised by y mole of MnO_4^- in acidic medium. Ratio of x/y is-

- (A) 2 : 1
(B) 10 : 1
(C) 3 : 1
(D) 1 : 3

7. 6×10^{-3} mole $K_2Cr_2O_7$ reacts completely with 9×10^{-3} mole X^{n+} to give XO_3^- and Cr^{3+} . The value of n is

- (A) 1 (B) 2
(C) 3 (D) None of these

8. Consider the reaction :



Pick **INCORRECT** statements

- (1) Oxidation number of Cr changes in the reaction
- (2) Cr is hexavalent in both reactant and product side
- (3) Covalency of Cr changes in the reaction
- (4) Total number of peroxy linkages on reactant and product side is same in the balanced reaction

- (A) 1,2 both (B) 2, 3 both
(C) 1, 3 both (D) 1,2,3,4 all

9. Which of the following is not perfect octahedral?

- (A) XeF_6
(B) SbCl_6^{-3}
(C) SiF_6^{-2}
(D) PCl_6^{-1}

10. Which of the following can be said to be NOT related to Drago's rule/application of Drago's rule.

- (A) Bond angle of PH_3 is roughly 94°
(B) NH_3 is easily soluble in water and forms NH_4^+ but PH_4^+ is not easily formed.
(C) H_2O is a better base than H_2S
(D) The axial bonds in PCl_5 are longer than equatorial bonds.

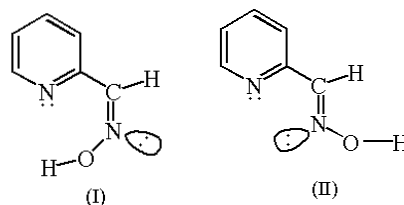
11. An element of 4th period having maximum number of unpaired electrons in it's ground state, forms a compound which shows a spin magnetic moment of 3.88 B.M. Find oxidation number of element in the compound.

- (A) 2 (B) 0 (C) 3 (D) 4

12. Which of the following facts is not related to inter-molecular H-bonding?

- (A) $\text{H}_2\text{O} \gg \text{H}_2\text{S}$ (B.P. order)
(B) Boric acid is a slippery solid at room temperature
(C) Acetic acid has dimer forming tendency when added to solvent benzene
(D) $n\text{-C}_5\text{H}_{12} > n\text{-C}_5\text{F}_{12}$ (B.P. order)

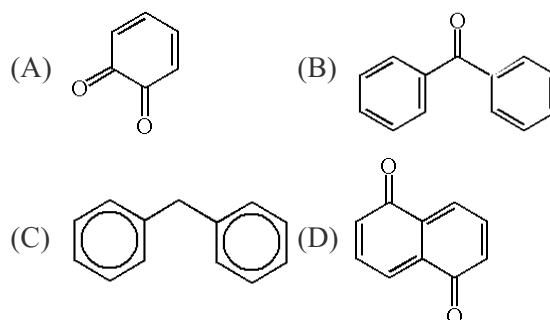
13. Consider the following two structure of same molecule :



Identify correct statement(s).

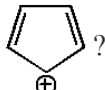
- (A) B.P. (I) > B.P. (II)
(B) B.P. (I) < B.P. (II)
(C) I > II (Viscosity)
(D) I < II (Volatile nature)

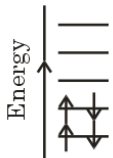
14. In which of the following molecules, all atoms are not coplanar?

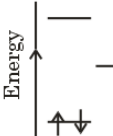


15. Which of the following statements is true about the contributions of canonical structures in the resonance hybrid?

- (A) A charge separated canonical structure contributes more than a canonical structure with out charge-separation
- (B) A canonical structure with less number of π bonds contributes more than a canonical structure with more number of π bonds
- (C) A canonical structure with negative charge on less electronegative atom contributes more than a canonical structure with negative charge on more electronegative atom
- (D) Contribution of a charge separated canonical structure will be more in a polar solvent than in a non polar solvent

16. Which of the following π -molecular orbital energy diagrams is correct for ?

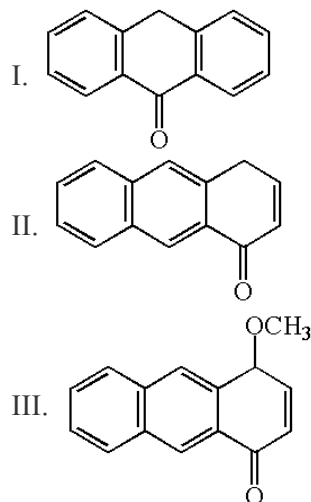
- (A) 

(C) 

(B) 

(D) 

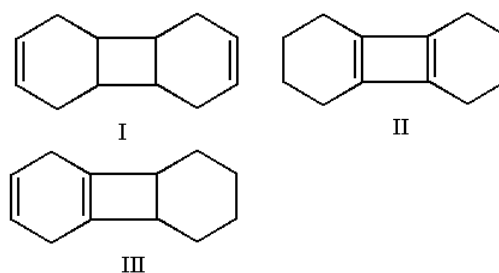
17.



Which among these can exhibit tautomerism?

- (A) I only
- (B) II only
- (C) III only
- (D) all of these

18.



Which of the following orders is correct for heat of hydrogenation of these compounds?

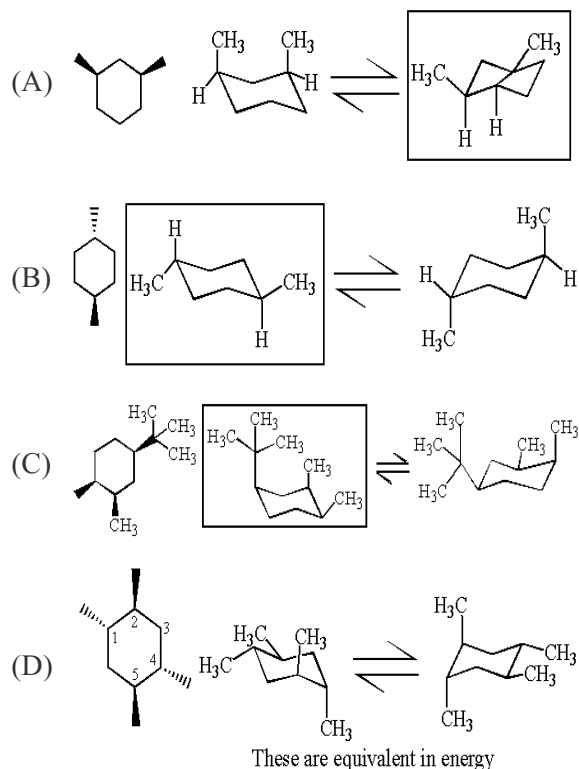
- (A) $I > II > III$
- (B) $III > II > I$
- (C) $III > I > II$
- (D) $II > III > I$

19. Match for the correct bondline structure.

I		P	
II		Q	
III		R	
IV		S	

- (A) I-R, II-P, III-S, IV-Q
 (B) I-P, II-R, III-S, IV-Q
 (C) I-R, II-P, III-Q, IV-S
 (D) I-S, II-P, III-S, IV-P

20. In which of the following the stable isomer is incorrectly labeled



SECTION-II : (Maximum Marks: 20)

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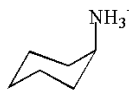
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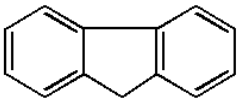
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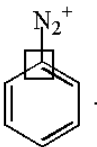
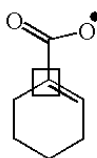
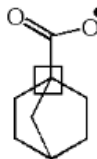
Zero Marks : 0 In all other cases.

1. Count the total number of unshared valence electrons on $I_3^- = x$
Count the total number of $p\pi - d\pi$ bonds in $PO_4^{3-} = y$
Count the total number of lone pairs in $XeF_4 = z$
Hence, find the value of $x - z - y$.
2. If in XeO_2F_2 , two oxygen atoms are replaced with fluorine atoms, then, number of properties which will be changed out of the following: (covalency of xenon remaining same)
 - a) Geometry of molecule
 - b) Bond angles
 - c) Hybridization of molecule
 - d) Number of lone pairs on Xenon
 - e) Total number of σ bonds.
3. Consider given electronic configuration of atom/ion.
 $1s^0 2s^0 2p^0 3s^0 3p^0 3d^0 4s^0 4p^1$
Find the total number of quantum number sets which are possible for electron given in the above configuration without changing its energy.
4. 500 ml of each three sample of H_2O_2 labelled 10 V, 15V and 20 V are mixed and then diluted with equal volume of water. The volume strength of resultant H_2O_2 solution is (Considering 1 mole of gas at STP = 22.7 L)
5. Find the number of moles of $KMnO_4$ needed to oxidise one mole of Cu_2S in acidic medium.
 $KMnO_4 + Cu_2S \rightarrow Mn^{+2} + Cu^{+2} + SO_2$
6. In acidic medium, dichromate ion oxidizes stannous ion as -
 $xSn^{2+} + yCr_2O_7^{-2} + zH^+ \rightarrow aSn^{+4} + bCr^{+3} + cH_2O$
where x, y, z, a, b and c are respective stoichiometric coefficients. Determine the value of $[(x + y + z) - (a + b + c)]$
7. A hydrocarbon C_xH_y on complete combustion gives $CO_2(g)$ & $H_2O(l)$. If volume of $O_2(g)$ required for complete combustion of hydrocarbon is double the volume of $CO_2(g)$ produced, then find the value of $\left(\frac{y}{x}\right)$.
8. How many of the following statements is/are true.
 1. No. of conformers of ethane are 3.
 2. No. of conformers of Hexadeuteroethane are infinite.
 3. The conformers of propane are degenerate isomers.
 4. The stable conformer of 1,2-dibromobutane is staggered form and the dipole moment of 1,2-dibromobutane is zero.
 5. 2-iodoethanol: gauche form is more stable than staggered because of H-bonding.
 6. CCl_4 : Do not show conformational isomerism.
 7.  more solvated than  in water.
 8. In basic medium staggered form of succinic acid (about C_2-C_3 axis) is more stable than its gauche form.
 9. In cyclohexane, potential energy : Boat form > Twist boat form > Chair form.

9. In how many options, carbanion intermediate is involved?

- (a) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{OH} \xrightarrow[\Delta]{\text{NaOH}}$
- (b)  $\xrightarrow{\text{KH}}$
- (c) $\text{H}_3\text{C}-\text{C}\equiv\text{CH} \xrightarrow{\text{NaNH}_2}$
- (d) $\text{H}_3\text{C}-\text{OH} \xrightarrow{\text{Na}}$

10. In how many cases, the hybridization of circled (indicated) carbon atom is same in both reactant and intermediate?

- (i) $\text{H}_3\text{C}-\text{Cl} \longrightarrow \text{CH}_3^+$
- (ii) $\text{H}_2\text{C}=\text{CH}-\text{Cl} \longrightarrow \text{H}_2\text{C}=\text{CH}^+$
- (iii) $\text{H}_2\text{C}=\text{CH}-\text{C}(=\text{O})-\text{O}^- \xrightarrow{\Delta} \text{H}_2\text{C}=\text{CH}^- + \text{CO}_2$
- (iv) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{O}^\bullet \longrightarrow \text{CH}_3^\bullet + \text{CO}_2$
- (v)  $\xrightarrow{\Delta} \text{Cyclohexadienyl}^+ + \text{N}_2$
- (vi)  $\longrightarrow \text{Cyclohexadienyl}^\bullet$
- (vii) $\text{HC}\equiv\text{C}-\text{Cl} \longrightarrow \text{HC}\equiv\text{C}^+$
- (viii) $\text{H}_2\text{C}=\text{C}=\text{O} \xrightarrow{\Delta} \text{singlet } \text{CH}_2 + \text{CO}$
- (ix)  $\longrightarrow \text{Cyclohexadienyl}^\bullet + \text{CO}_2$

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- Let A(1,3), B(a,b), C(4,7) are collinear points and $\sqrt{(a-1)^2 + (b-3)^2} + \sqrt{(4-a)^2 + (7-b)^2} = k$, then minimum value of 'k' is -
(A) 4
(B) 5
(C) 6
(D) 9
- If points A(8, 3), B(a, 5), C(85, 25) are collinear, then value of a is -
(A) 12 (B) 15
(C) 18 (D) 22
- If A(1, 0), B(-1, 0) and C(2, 0) then the locus of point P satisfying $PB^2 + PC^2 = 2AP^2$ is -
(A) $x = 1$
(B) $x = 2$
(C) $y = 1$
(D) $y = 2$

- If the numbers $\frac{1}{k+2}$, $\frac{1}{k^2}$ and $\frac{1}{4k+5}$ are three consecutive terms of an H.P., then number of values of k is
(A) 1
(B) 2
(C) 3
(D) 4
- The sum of infinite terms of the series :-
 $\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \frac{1}{3.4.5.6} + \dots$
(A) 1
(B) $\frac{1}{24}$
(C) $\frac{1}{18}$
(D) $\frac{1}{30}$
- If $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (x-y)(y-z)(z-x) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$,
(x, y, z are non-zero distinct real number), then n is equal to :-
(A) -2
(B) -1
(C) 0
(D) 1

7. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$ has a non-trivial solution, then both the roots of equation $at^2 + bt + c = 0$ are -

(A) 0
(B) positive
(C) negative
(D) of opposite sign

8. The system of linear equations

$$x + 3y + 2z = -2$$

$$x + 3y + 4z = -4$$

$$x + 3y = 0$$

(A) unique solution
(B) infinite solution
(C) no solution
(D) unique trivial solution

9. Let x, y are integers such that $81^y - 4^x = 17$, then $2x + 3y$ is -

(A) 4
(B) -3
(C) 2
(D) 9

10. The value of $\begin{vmatrix} 1 & x & y \\ 2 & \sin x + 2x & \sin y + 2y \\ 3 & \cos x + 3x & \cos y + 3y \end{vmatrix}$ is

(A) $\cos(x + y)$ (B) $\cos(xy)$
(C) $\sin(x + y)$ (D) $\sin(x - y)$

11. If a, b & c are unequal natural numbers, then

$$\begin{vmatrix} a^2 + b^2 & ab + bc & ac + ba \\ ba + cb & b^2 + c^2 & bc + ca \\ ca + ab & cb + ac & c^2 + a^2 \end{vmatrix}$$
 is equal to -

(A) $a + b + c$
(B) $a + b$
(C) $b + c$
(D) 0

12. If $\log_2 x = \log_7 8$, $\log_3 y = \log_5 9$, $\log_5 z = \log_{11} 25$, then the value of $x^{(\log_2 7)^2} + y^{(\log_3 5)^2} + z^{(\log_5 11)^2}$ is equal to -

(A) 389
(B) 489
(C) 589
(D) 289

13. Two real numbers α & β are such that $\alpha + \beta = 3$ & $|\alpha - \beta| = 4$, then α & β are the roots of the quadratic equation :

(A) $4x^2 - 12x - 7 = 0$
(B) $4x^2 - 12x + 7 = 0$
(C) $4x^2 - 12x + 25 = 0$
(D) none of these

14. If the area of $\Delta(ABC)$ is 25 square units, then the value of $c^2 \sin 2B + b^2 \sin 2C$ is - (where symbols have their usual meaning)

(A) 25 (B) 50
(C) 75 (D) 100

15. If s is semi-perimeter of a triangle ABC (usual notations), then the value of $\left(\sin^2 \frac{B}{2} + \sin^2 \frac{A}{2} + c\right)$ is :-

(A) s (B) $\frac{s}{2}$
(C) $2s$ (D) $s - c$

16. The solution of $\frac{1}{2} + \cos x + \cos 2x + \cos 3x + \cos 4x = 0$ is -

(A) $x = \frac{2n\pi}{9}, n \in I, n \neq 9m, m \in I$
(B) $x = \frac{2n\pi}{9}, n \in I, n = 9m, m \in I$
(C) $x = \frac{n\pi}{9} + \frac{\pi}{2}, n \in I$
(D) $x = \frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$

17. If $\sin^4 x + \cos^4 x + \sin 2x + \alpha = 0$ has a solution, then number of integral values of α is/are ($\alpha \in R$)

(A) 0 (B) 2
(C) 4 (D) 3

18. Roots of $x^3 + ax^2 + bx + c = 0$ are cosines of angles of an acute triangle, then the value of $a^2 - 2b - 2c$ is

(A) 0 (B) 1
(C) -1 (D) 3

19. Product of the solutions of the equation $|\log_{\sqrt{3}} x - 2| - |\log_3 x - 2| = 2$ is

(A) -1 (B) 3
(C) 2 (D) 1

20. Complete set of values of k for which the equation $4^x - (k + 2)2^x + 2k = 0$, has exactly one positive root

(A) R (B) $(-\infty, 1]$
(C) $\left(\frac{1}{2}, \infty\right)$ (D) $(0, \infty)$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Number of solutions of $x \in [0, \pi]$ satisfying the equation $(\log_{\sqrt{3}} \tan x) \left(\sqrt{\frac{\log_{\sqrt{3}} (\sqrt{3})^3}{\log_{\sqrt{3}} \sqrt{3}}} + \frac{\log_{\sqrt{3}} 3}{\log_{\sqrt{3}} \tan x} \right) = -1$ are
2. Number of solution of $\cos x \cos 8x = -1$ in $[0, 5\pi]$ is
3. Sum of possible values of k for which roots of $x^2 + (k + 1)x + \lambda = 0$ are square of each other

4. If $\begin{vmatrix} 1 & x & x^2 \\ x & x^2 & 1 \\ x^2 & 1 & x \end{vmatrix} = 3$, then value of $\begin{vmatrix} x^3-1 & 0 & x-x^4 \\ 0 & x-x^4 & x^3-1 \\ x-x^4 & x^3-1 & 0 \end{vmatrix}$ is
5. In ΔABC given that $9a^2 + 9b^2 - 17c^2 = 0$. If $\frac{\cot A + \cot B}{\cot C} = \frac{m}{n}$ where m and n are relatively prime, then value of $m - n$ equal to
6. If A, B, C are the angles of a triangle such that $\cos^2 A + \cos^2 B + \sin^2 C - \sqrt{2}\cos A - \sqrt{2}\sin C + 1 = 0$, then value of $\sin^2 A + \sin^2 B + \cos^2 C$, is
7. For what value of k to the following system of equations possess a non-trivial solution ?
 $x + ky + 3z = 0$
 $3x + ky - 2z = 0$
 $2x + 3y - 4z = 0$
8. Sum of all possible integral value(s) of x for which $|1 - x|^{\log_2(x+1)} > |x - 1|^{\log_2(4-x)}$ is
9. If the area of the circle which is inscribed in an equilateral triangle of side length ' ℓ ' is $\frac{\pi \ell^2}{4\lambda}$, then λ is
10. The number of integer values of k for which the inequality $k\cos^2 x - k\cos x + 1 \geq 0 \forall x \in \mathbb{R}$.

Space for Rough Work

SPACE FOR ROUGH WORK

